LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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We give a negative answer to the longstanding question whether every countable group is MF. Here MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. Our main tool is the following one-sided compression criterion. Let G be countable, let $L \leq G$ have property (T), and let $K \trianglelefteq G$ have property (T). If K is contained in the normal subgroup generated by the commutators $[ucu^{-1}, \ell]$, where $uLu^{-1} \leq L$, $c \in C_G(L)$, and $\ell \in L$, then every homomorphism from G to an MF group kills K. For $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$, a single commutator of this form normally generates H. The group H is nontrivial, simple, and has property (T), whereas every homomorphism from H to an MF group is trivial. In particular, H is not MF. Consequently, the reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF. Finally, if $f\colon G \to Q$ is surjective and $\ker f \leq \mathrm{Rad}_{\mathrm{MF}}(G)$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q))$. In this case, image and inverse image identify the normal subgroups $N \trianglelefteq G$ satisfying $\ker f \leq N$ and G/N MF with the normal subgroups $P \trianglelefteq Q$ satisfying Q/P MF.

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1. Introduction

Not every countable group is MF. Our counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

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and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

Thus the models become multiplicative in operator norm without losing any nonidentity element. This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum.

A separable C^*-algebra is called MF if it embeds as a C^*-subalgebra of a norm matrix corona. For every countable group G, the reduced group algebra $C^*_{\mathrm{r}}(G)$ is separable and has a faithful canonical trace, hence is stably finite. If $C^*_{\mathrm{r}}(G)$ is MF, restricting an MF embedding to its canonical group unitaries shows that G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^*-algebra that is not MF.

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

Proposition 1.1 shows that G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF kernel closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap \{\ker f : N \le \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 1.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism of G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R as $(x_j)_{j \ge 1}$ and the pairs in $(G/R)^2$. For each x_j, the definition of R supplies a corona homomorphism whose value at x_j has positive distance from the identity. Every such homomorphism kills R and therefore descends to G/R. Choose coordinate unitary lifts, using polar correction as in Lemma 4.1. At stage n, take a direct sum of one coordinate from each of the first n models. Choose each coordinate far enough out that the first n multiplication defects are at most $1/n$ and the designated value at x_j retains at least half of its corona norm separation. Such coordinates exist arbitrarily far out: the defects converge to zero, while the quotient norm is a limsup. The resulting operator norm asymptotic representation detects every x_j, so its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all maps into MF groups that kill N is N, which is the asserted closure criterion. $\square$

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) *and $\mathfrak{D}_G(L) = G$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.*

We sketch the argument. Corollary 3.4 gives $\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G)$. Equivalently, if (V_n) is any operator norm asymptotic representation of G, then $\|V_n(d) - 1\|_2 \to 0$ for every $d \in \mathfrak{D}_G(L)$. To upgrade this conclusion to corona triviality on K, fix a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ and suppose, toward a contradiction, that Θ is nontrivial on K. Let p be the image in $\mathcal{Q}_{\mathbf{d}}$ of the Kazhdan projection of K. Because $K \trianglelefteq G$, the projection p commutes with $\Theta(G)$; hence $q = 1 - p$ defines a G-invariant complementary corner. Lemma 4.1 realizes the action on this corner by an operator norm asymptotic representation (W_n) on matrix corners with identities q_n. The corner has no nonzero K-fixed vectors, so the Kazhdan inequality yields a single $s_0 \in K$ and a subsequence along which $\|W_n(s_0) - q_n\|_2$ stays bounded below by a positive constant. This contradicts $s_0 \in K \leq \mathfrak{D}_G(L)$ and the preceding convergence. Thus every corona homomorphism kills K, as required.

Theorem 2.1 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the binary Leavitt construction that produces the group H announced above.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1t_1)s_1 = 1$ and the elementary commutator relations show directly that d normally generates H. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 3.3.

A full MF radical also lets us prescribe the quotient visible to MF.

Theorem C (prescribed quotients visible to MF)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q\colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. The operator-algebraic lineage of the problem predates the term *MF group*. In a 1994 Oberwolfach abstract, Blackadar and Kirchberg suggested that every separable nuclear stably finite C^*-algebra might be NF [BK94]. Their 1997 paper introduced MF C^*-algebras; for separable nuclear algebras, the MF and NF properties are equivalent [BK]. They also observed that no separable stably finite C^*-algebra was known not to be MF. The broader question whether every separable stably finite C^*-algebra is MF became known as the Blackadar–Kirchberg MF problem [RS].

Carrión–Dadarlat–Eckhardt later introduced the precise group property used here [CDE]. Subsequent approximation literature attributed the question whether every countable group is MF to Kirchberg [LO, Th]. The negative solution of the Connes embedding problem also produced stably finite non-MF C^*-algebras in general [JNVWY]. Theorem B provides such an example among reduced group C^*-algebras and answers the operator norm group approximation question negatively. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves MF permanence for doubles $G *_H G$ of MF groups. Separately, Shulman proves MF full group C^*-algebra results for several classes of amalgamated and semidirect products [Sh]. The operator norm case remained open after failure of approximation had been established for finite Schatten norms [DGLT, LO, Th]. Bachner–Dogon–Lubotzky study stability between the operator and Hilbert–Schmidt norms as a possible source of non-MF groups [BDL]. Here operator norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. The Kazhdan projection of this conjugation representation makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Theorem 4.2 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras $L_K(1,n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's work gives general theorems about normal subgroups of linear groups over exchange rings [Pre]; the simplicity proof below instead extracts an elementary root directly from a nonidentity normal element. Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1].

Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \le G$, and suppose $u \in G$ satisfies $uLu^{-1} \le L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. □

The finite-dimensional hypothesis is essential here. On an infinite-dimensional space an injective endomorphism of the commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity for projections onto fixed vectors.

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator norm asymptotic representations of G.

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 3.1. *For every sequence (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^* x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^* x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1-q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. $\square$

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. $\square$

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2 \|A - B\|,$$

so the operator norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\mathrm{Fix}\,\sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \le L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. ☐

Corollary 3.4. *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 3.3, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\left\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \right\| \longrightarrow 0.$$

Therefore

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. Normal Kazhdan subgroups and the MF radical

Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona representation of G. Normality of K implies that p commutes with the image of G. Hence $q = 1 - p$ also commutes with that image, and the corner $q\mathcal{Q}_{\mathbf{d}}q$ has no nonzero K-fixed vectors.

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator norm asymptotic representation. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates. □

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable, let $D \leq R_{\infty\to 2}(G)$, and let $K \trianglelefteq G$ have property* (T) *with $K \leq D$. Then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\operatorname{Fix}\Theta(K)$ and $\operatorname{Fix}\Theta(K)$; these agree because $gKg^{-1}=K$. Put $q=1-p$. The projection q is nonzero: if $p=1$, then every vector is fixed by K, so every element of K has trivial corona image. Lemma 4.1 gives an operator norm asymptotic representation (W_n) on nonzero corners $q_nM_{d_n}(\mathbb{C})q_n$.

Choose a finite symmetric Kazhdan set $S\subseteq K$ and a Kazhdan constant $\kappa>0$. In the corner $q\mathcal{Q}_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b=\frac{1}{|S|}\sum_{s\in S}(q\Theta(s)q-q)^*(q\Theta(s)q-q)$$

satisfies

$$b\geq\frac{\kappa^2}{|S|}\,q.$$

Indeed, in every representation of the corner there are no K-fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n=\frac{1}{|S|}\sum_{s\in S}(W_n(s)-q_n)^*(W_n(s)-q_n)$$

represent b. Taking the normalized trace on $q_nM_{d_n}(\mathbb{C})q_n$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s)-q_n\|_2^2\geq\frac{\kappa^2}{|S|}-o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n-(\kappa^2/|S|)q_n$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0\in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0\in K\leq D\leq R_{\infty\to2}(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u\in\operatorname{Comp}_G(L)$, $c\in C_G(L)$, and $\ell\in L$, Corollary 3.4 gives

$$[ucu^{-1},\ell]\in R_{\infty\to2}(G).$$

Normality of $R_{\infty\to2}(G)$ therefore gives

$$\mathfrak{D}_G(L)\leq R_{\infty\to2}(G).$$

Theorem 4.2 now gives $K\leq\operatorname{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K=G=\mathfrak{D}_G(L)$ gives $\operatorname{Rad}_{\mathrm{MF}}(G)=G$.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. □

Proposition 4.3 (functoriality and normal generation)**.** *Let $L \leq G$ and put $D = \mathfrak{D}_G(L)$.*

If $f\colon G \to Q$ is a homomorphism, then

$$f(D) \leq \mathfrak{D}_{f(G)}(f(L)). \tag{4}$$

If $S \leq G$ is a nontrivial simple subgroup and $D \cap S \neq 1$, then $S \leq D$. If in addition the normal closure of S in G is all of G, then $D = G$. If moreover G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = G.$$

Alternatively, suppose that f is onto, $S \leq D$ is simple, $f(S) \neq 1$, and $f(S)$ normally generates Q. Then

$$\mathfrak{D}_Q(f(L)) = Q.$$

If, in addition, G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(Q) = Q.$$

Proof. If $u \in \mathrm{Comp}_G(L)$, then $f(u)f(L)f(u)^{-1} \leq f(L)$. If $c \in C_G(L)$, then $f(c) \in C_{f(G)}(f(L))$. Therefore f takes every defining generator of D into a defining generator of $\mathfrak{D}_{f(G)}(f(L))$. This proves (4).

The subgroup D is normal in G, so $D \cap S$ is normal in S. Simplicity and $D \cap S \neq 1$ give $D \cap S = S$, hence $S \leq D$. Since D is normal in G, it then contains the normal closure of S. When the normal closure of S is G, this proves $D = G$. Under the property (T) hypotheses, Theorem A with $K = G$ then gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

For the image statement, (4) gives $f(S) \leq \mathfrak{D}_Q(f(L))$. The latter subgroup is normal in Q, so it contains the normal closure of $f(S)$ and is therefore all of Q. Property (T) passes to quotients, so the final hypotheses give property (T) for both $f(L)$ and Q. Applying Theorem A inside Q, with $K = Q$, proves the last assertion. ☐

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{5}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{6}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $qs_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$

proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_3 to pI_3.

We now realize the algebraic compression $A \mapsto \Psi(A)$ as conjugation by an elementary matrix. The following matrices provide the required change of coordinates. Set

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$\tau = \operatorname{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{7}$$

The purpose of τ is the identity

$$\tau \operatorname{diag}(A, I_9)\, \tau^{-1} = \operatorname{diag}(\Psi(A), I_9). \tag{8}$$

In fact, τ lies in the elementary group, not merely in the general linear group.

Elementary membership of τ. Writing $I = I_6$, the Whitehead factorization is

(9)

$$\operatorname{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

Each factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block can be inserted one entry at a time because the corresponding matrix units have pairwise zero products. Since $Y = X^{-1}$, (9) proves $\tau \in \mathrm{EL}_{12}(R)$. $\square$

Let $H = \mathrm{EL}_{12}(R)$ and let $L = \mathrm{EL}_3(R) \leq H$ be the corner in the upper left. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ, Theorem 1.1]. For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$\tau L \tau^{-1} \leq L. \tag{10}$$

The field $\mathbb{F}_2$ is used in the diagonal branch of the simplicity argument below: the only central unit of R is then 1. The dimension 12 comes from applying the 6×6 Whitehead factorization to the 3×3 Kazhdan corner.

6. The binary Leavitt group has full MF radical

Proposition 6.1. *The group $H = \mathrm{EL}_{12}(R)$ is simple.*

Proof. We use two concrete ring facts. First, for every $0 \neq x \in R$ there are $a, b \in R$ such that

$$axb = 1. \tag{11}$$

This is the form of pure infiniteness needed here [AA]. Second, the only central unit of R is 1: indeed $Z(R) = \mathbb{F}_2$ by [AC, Corollary 4.3] and $\mathbb{F}_2^\times = \{1\}$.

Normal generation from one root. We first record that a nonzero root normally generates the elementary group. More generally, let S be a unital ring, let $n \geq 3$, and suppose $a, x, b \in S$ satisfy $axb = 1$. If a normal subgroup of $\mathrm{EL}_n(S)$ contains $e_{ij}(x)$, choose k different from i and j. The elementary commutator relations give

$$[e_{ij}(x), e_{jk}(b)] = e_{ik}(xb), \qquad [e_{ji}(a), e_{ik}(xb)] = e_{jk}(1),$$

so the normal subgroup contains a unit root. From any $e_{uv}(1)$ in the normal subgroup and any third index w, the relations

$$[e_{wu}(r), e_{uv}(1)] = e_{wv}(r), \qquad [e_{uv}(1), e_{vw}(r)] = e_{uw}(r)$$

show that the normal subgroup contains the corresponding source and target moves for every coefficient $r \in S$. Repeating these moves, using a third index when needed, gives every elementary generator. Thus, by (11), it remains to prove that every nontrivial normal subgroup $N \trianglelefteq H$ contains a nonzero elementary root.

Separating two coefficients. We also need the following coefficient-separation consequence of the Leavitt relations. Its purpose is to make one matrix product vanish while the reverse product retains a nonzero entry. If $r, s \in R$ are nonzero, then there are $a, b \in R$ such that

$$arb = 0, \qquad bsar \neq 0. \tag{12}$$

For every $w \in R$ there are $x, y \in R$ with $xwy = 0$ and $yx \neq 0$. This is immediate with $x = y = 1$ if $w = 0$. Otherwise choose c, d with $cwd = 1$ and put $x = t_1 c$, $y = ds_0$. Then $xwy = t_1 s_0 = 0$. Also $t_0(s_0 t_1)s_1 = 1$, so $s_0 t_1 \neq 0$. If $yx = ds_0 t_1 c$ were zero, multiplying this equality on the left by cw and on the right by wd, and using $cwd = 1$, would force $s_0 t_1 = 0$. Therefore $yx \neq 0$. To obtain (12), we choose four sandwich factorizations. They serve one bookkeeping purpose: if the desired reverse product $bsar$ were zero, the outer factors below would reduce that equality to the contradiction $yx = 0$.

$$c_r\, r\, d_r = 1, \quad e_d\, d_r\, f_d = 1, \quad e_s\, s\, f_s = 1, \quad e_t(c_r r) f_t = 1,$$

Apply the preceding construction to $w = e_t f_d$. Set

$$a = f_s x e_t c_r, \qquad b = d_r f_d y e_s.$$

Then $arb = 0$. If $bsar = 0$, multiplication on the left by e_d and on the right by f_t gives $yx = 0$, a contradiction.

Extracting a root from a normal subgroup. Now let $N \trianglelefteq H$ be nontrivial and choose $1 \neq g \in N$. Suppose first that g is diagonal. If g commuted with every $e_{ij}(a)$, then commuting with the $e_{ij}(1)$ would force $g = \lambda I_{12}$. Commuting with arbitrary $e_{ij}(a)$ would then make λ a central unit. Hence $g = 1$, a contradiction. Thus for some $i \neq j$ and $a \in R$, conjugation by g sends $e_{ij}(a)$ to $e_{ij}(y)$ with $y \neq a$. Consequently

$$[g, e_{ij}(a)] = e_{ij}(y - a) \in N$$

is a nonzero root.

Suppose next that g is not diagonal, and choose $\ell \neq i$ with $g_{\ell i} \neq 0$. In both remaining cases, the goal is to manufacture an element $1+v \in N$ whose nonzero square-zero defect v is supported in one row; one final commutator will then extract a nonzero elementary root. Put $r = (g^{-1})_{\ell i}$.

If $r = 0$, set

$$A = gE_{i\ell}g^{-1}, \qquad B = E_{i\ell}.$$

Then $AB = 0$, and the double commutator

$$[ge_{i\ell}(1)g^{-1}, e_{i\ell}(1)] = 1 - BA \tag{13}$$

belongs to N. Its defect $-BA$ is square-zero and supported in row i. There is an m for which $g_{\ell i}(g^{-1})_{\ell m} \neq 0$: otherwise the (ℓ, ℓ) entry of $g^{-1}g = 1$ would give $g_{\ell i} = 0$. Thus the (i, m) entry of the defect in (13) is nonzero.

If $r \neq 0$, apply (12) to r and $s = g_{\ell i}$, obtaining a, b with $arb = 0$ and $bsar \neq 0$. This time put

$$A = g(aE_{i\ell})g^{-1}, \qquad B = bE_{i\ell}.$$

Again $AB = 0$, because its potentially nonzero coefficient contains arb. As the two elementary roots commute, reduction modulo N shows that

$$[ge_{i\ell}(a)g^{-1}, e_{i\ell}(b)] = 1 - BA \in N. \tag{14}$$

The defect is square-zero and supported in row i, while its (i, i) entry is $-bsar \neq 0$.

Both off-diagonal cases therefore produce an element $1 + v \in N$ with $v^2 = 0$. The matrix v is supported in a single row i and has some nonzero entry v_{im}. Choose n different from both i and m (with only $n \neq i$ needed when $m = i$). Direct multiplication gives the identity that extracts one row,

$$[1 + v, e_{mn}(1)] = e_{in}(v_{im}) \in N.$$

This is a nonzero root, so the argument about normal generation above gives $N = H$. Hence every normal subgroup of H is trivial or all of H. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c\tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. Every elementary generator of the 3×3 corner in the upper left has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that each such generator commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (7) gives

$$\tau c\tau^{-1} = e_{01}(q)e_{34}(1). \tag{15}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since

$t_1 q s_1 = 1$, the normal-generation argument following (11) applied to $e_{02}(q)$ gives $\langle\langle d \rangle\rangle_H = H$. $\square$

Proof of Theorem B. The finitely generated $\mathbb{F}_2$-algebra R is countable, and hence so is H. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (10), the centrality of c relative to L, and Proposition 6.2 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 6.2 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Independently of this radical calculation, Proposition 6.1 shows that H is nontrivial and simple.

If M is MF and $f\colon H \to M$ is a homomorphism, compose f with a faithful corona embedding of M. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C^*_{\mathrm{r}}(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If $C^*_{\mathrm{r}}(H)$ were MF, restricting an MF embedding to the canonical unitaries would make H an MF group, a contradiction. $\square$

7. Quotients visible to MF

The MF radical is exact across every surjection whose kernel is already invisible to MF.

Proposition 7.1 (full-kernel pullback)**.** *Let $f\colon G \to Q$ be a surjective homomorphism of countable groups. If $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

In particular, the conclusion holds whenever $\mathrm{Rad}_{\mathrm{MF}}(\ker f) = \ker f$. More generally, for every normal subgroup $N \trianglelefteq G$,

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = f^{-1}\left(\mathrm{cl}^Q_{\mathrm{MF}}(f(N))\right).$$

Consequently,

$$G/N \text{ is MF} \iff \ker f \le N \text{ and } Q/f(N) \text{ is MF.}$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f. Conversely, composing a corona homomorphism of Q with f gives one of G. Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the final assertion, functoriality of the MF radical under the inclusion $\ker f \to G$ gives $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N, gives the closure formula. If this closure equals N, then it contains $\ker f$. Mapping the equality onto Q shows that $\mathrm{cl}^Q_{\mathrm{MF}}(f(N)) = f(N)$. Conversely, these two conditions give

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 1.1, applied in G and Q. $\square$

Thus image under f and inverse image under f give mutually inverse correspondences, preserving inclusion between the normal subgroups with MF quotient in G and in Q. The map induced by f on the largest quotients visible to MF is an isomorphism:

$$G/\operatorname{Rad}_{\mathrm{MF}}(G) \cong Q/\operatorname{Rad}_{\mathrm{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with kernels invisible to MF.

Let B be a countable group with $\operatorname{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{16}$$

where $A \to Q \times A$ is the inclusion into the second factor. The trivial map $B \to Q$ and the projection $Q \times A \to Q$ agree on A. They therefore define an epimorphism

$$\pi_Q \colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (16) are injective; in particular, $d \neq 1$ in W_Q.

Proposition 7.2 (universal factorization)**.** *Let T be a group such that every homomorphism $B \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\operatorname{Hom}(Q, T) \longrightarrow \operatorname{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f \colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1) f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills B, since $\langle\langle d \rangle\rangle_B = B$, and kills the A-factor of $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. $\square$

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\operatorname{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.2 therefore gives the asserted bijection for every MF group M. In addition, the definition of $\operatorname{Rad}_{\mathrm{MF}}(B) = B$ says directly that every homomorphism from B to $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is trivial. Applying

Proposition 7.2 with $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ shows that every corona homomorphism from W_Q factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

□

If $N \trianglelefteq W_Q$, the same factorization gives

$$\mathrm{cl}_{\mathrm{MF}}^{W_Q}(N) = \pi_Q^{-1}\left(\mathrm{cl}_{\mathrm{MF}}^{Q}(\pi_Q(N))\right). \tag{17}$$

Indeed, a homomorphism from W_Q to an MF group kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (17) give

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \leq N$, then $N = \pi_Q^{-1}(\pi_Q(N))$.

Taking $Q = \mathbb{Z}$ makes the quotient criterion especially concrete: for every normal subgroup $N \trianglelefteq W_{\mathbb{Z}}$,

$$W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N. \tag{18}$$

Indeed, every quotient of $\mathbb{Z}$ is MF, while $\ker \pi_{\mathbb{Z}} = \langle\langle d \rangle\rangle_{W_{\mathbb{Z}}}$.

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